

Manthan Bhala

Machine Learning Engineer & Software Developer

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EXPERIENCE

Senior Machine Learning Engineer — Makro PRO

Oct 2023 – Present

Remote

- Developed e-commerce recommendation models that increased total basket additions by 52%.
- Built personalisation models by distilling LLMs into LightGBM for production-scale inference.
- Built a vector search system for e-commerce with p99 latency under 100ms using vLLM.
- Improved search results through autocategorization, autocorrect, query expansion, and query rewriting models.
- Built a chatbot POC with RAG pipeline and function calling, optimized with quantization and vLLM deployment.

ML Engineer — Frinks AI

Mar 2023 – Jul 2023

Bengaluru, Karnataka, India

- Developed and optimized ML inference pipelines for real-time defect detection and counting on industrial production lines.
- Built computer vision and backend services for a quality inspection platform for dimensioning, surface defect, and crack detection.

Data Scientist — Jaguar Land Rover

Sep 2020 – Feb 2023

Bengaluru, Karnataka, India

- Developed and maintained ML pipelines and backend services for Vehicle Algorithmic Self Diagnostics Tool.
- Worked on Advanced Driver Assistance Systems (ADAS) for autonomous and assisted driving features.

SKILLS

ML Engineering

- Python
- PyTorch
- TensorFlow
- Hugging Face
- LightGBM
- PEFT

Applied AI

- Information Retrieval
- Learning to Rank
- Recommendation Systems
- LLMs
- RAG
- Vector Search
- Computer Vision
- NLP

Data & Infrastructure

- SQL
- Docker
- Kubernetes
- CI/CD
- Redis
- vLLM

EDUCATION

Bachelor of Technology — Indian Institute of Technology (IIT), Bombay

2016 – 2020 | CGPA: 8.48/10

PATENTS

2501007631 — A Hybrid Lexical-Semantic Retrieval Process for E-Commerce Search

2501004476 — A Hybrid LLM Process for Multi-Objective Reranking in E-Commerce

2501004419 — A Multilingual Phonetic BK-Tree Based Spelling Correction Process in E-Commerce